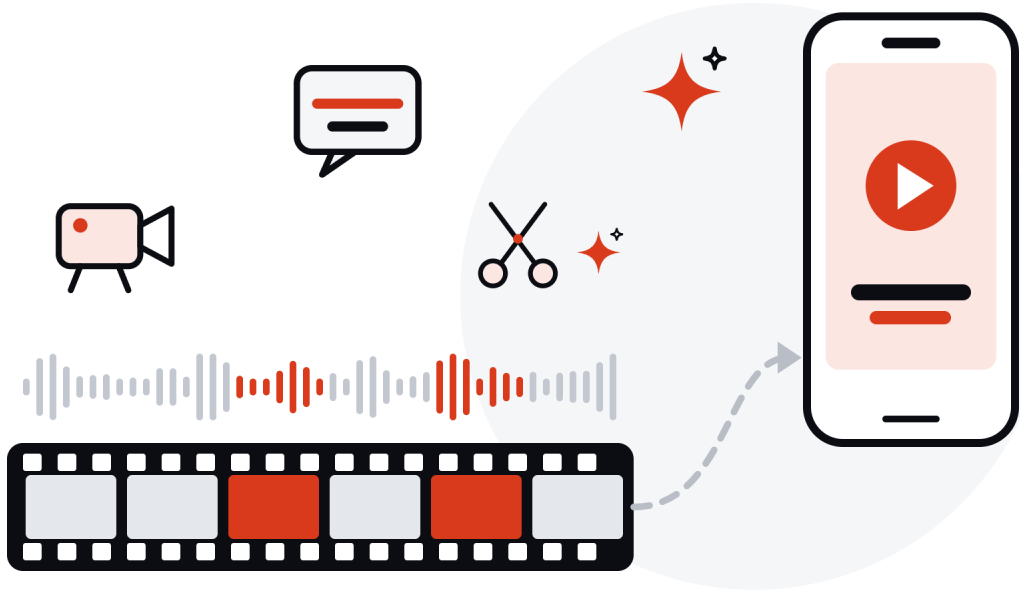
● REC · MILESTONE 2

# [SynopStudio]
# MS2

Video editing and summarization with artificial intelligence.

PEI — PROJETO EM ENGENHARIA INFORMÁTICA
· [2026/2027]




GROUP [15]

| | |
|---|---|
| [Daniel Duque] | [124880] |
| [David Monteiro] | [125793] |
| [Lucas Reis] | [126287] |
| [Murilo Frezzato] | [125487] |
| [Tomas Lopes] | [125596] |

SUPERVISORS

[Diogo Gomes]
[Rui Raposo]

deti
universidade de aveiro
departamento de eletrónica, telecomunicações e informática

SYNOPSTUDIO · IN ONE SENTENCE

# Long talks turned into sharp clips that respect your voice.

MILESTONE 2 · ELABORATION

# State of the art

✓ Supported ~ Partial / limited × Not supported

| | OpusClip | CapCut | Canva | Descript | Vizard | Kapwing | SynopStudio |
|---|---|---|---|---|---|---|---|
| Long Video Summarization (Auto-Clipping) | ✓ | ✓ | ~ | ~ | ✓ | ~ | ✓ |
| Professional Timeline and Multi-Track Editing | × | ✓ | ~ | ✓ | ~ | ✓ | ✓ |
| Automated Subtitling | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Dynamic Subtitles and Keyword Highlighting | ✓ | ✓ | × | ~ | ✓ | ~ | ✓ |
| Transcript Based Editing | ~ | ✓ | × | ✓ | ~ | ~ | ✓ |
| Speaker Tracking and Vertical Reframing | ✓ | ✓ | × | ~ | ✓ | ~ | ✓ |
| Import of Auxiliary Contextual Documents (Slides, Articles, Reports) | × | × | × | × | × | × | ✓ |
| Verifiable Source Traceability and Internal Linkage | × | × | × | × | × | × | ✓ |
| Context Loss Warnings | × | × | × | × | × | × | ✓ |
| Visual Indicator of Low Confidence (Transcripts, Generated Text, Speakers) | × | × | × | × | × | × | ✓ |
| Fully Open-Source Model Stack | × | × | × | ~ | × | ~ | ✓ |

# Five personas



## Eduardo

47 · Professor and researcher

WANTS

Short videos from his talks, reviewed and approved before publishing.



## António

52 · City councilor

WANTS

Clips that link to the exact moment in the original recording.

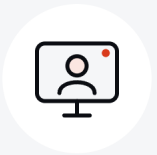


## João

24 · Streamer and content creator

WANTS

The best moments of each stream, exported vertical and subtitled.

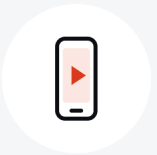


## Luciano

21 · Member of student association

WANTS

Suggested highlights he can adjust without learning a video editor.



## Paula

42 · Lawyer

WANTS

Short family clips, shared only with family and friends.

# Actors

### Content Creator

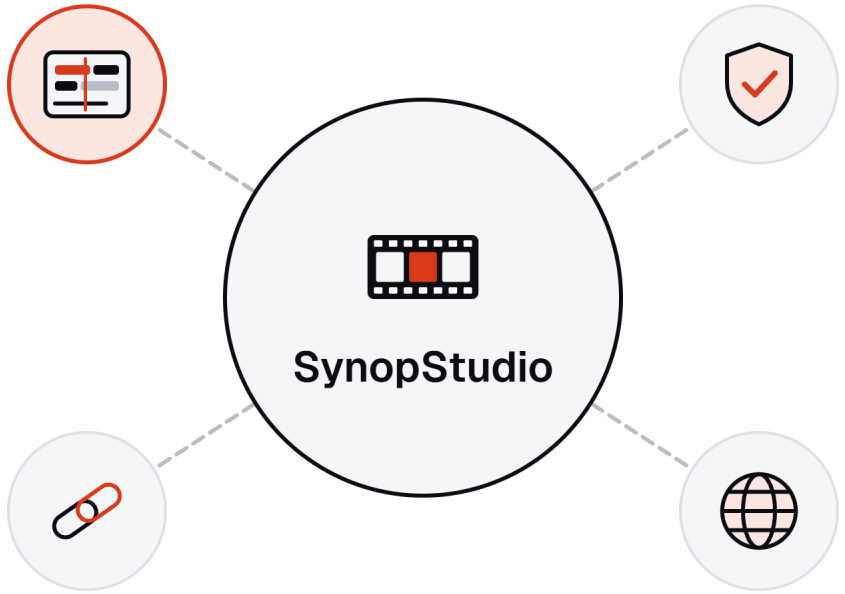
PRIMARY USER

Uploads videos and documents, reviews AI proposals, edits, approves and exports.

### Administrator

PLATFORM OPERATOR

Manages accounts, privacy and moderation rules, and audits the logs.




### Viewer

NO ACCOUNT

Opens a shared clip and checks it against the original moment.

### External Platform

SYSTEM

Social networks and video repositories, connected through the API.

# Use cases

Shown as flows at the end

| | | | |
|---|---|---|---|
| UC01 CREATOR | Create Project | UC02 CREATOR | Upload Local Video |
| UC04 CREATOR | Attach Reference Documents | UC05 CREATOR | Transcribe Video |
| UC08 CREATOR | Search Content in Natural Language | UC09 CREATOR | Configure Clip Generation Parameters |
| UC10 CREATOR | Generate Clip Proposals | UC11 CREATOR | Review Clip Proposal |
| UC12 CREATOR | Edit Clip via Transcript | UC15 CREATOR | Approve Clip for Export |
| UC22 ADMIN | Consult Audit Logs | UC25 VIEWER | Verify Clip Against Original Recording |

# Functional requirements

| | | |
|---|---|---|
| FR01 | Metadata, authorship and licensing registration | UC01 |
| FR02 | Multi-source long-form video ingestion | UC02 |
| FR03 | Ingestion of authorized auxiliary context documents | UC04 |
| FR04 | Synchronized audio transcription and time alignment | UC05 |
| FR05 | Speaker identification and voice turn attribution | UC05 |
| FR08 | Natural language semantic search | UC08 |
| FR09 | Summarization and output configuration | UC09 |
| FR10 | Automated clip generation focused on rigor | UC10 |
| FR11 | Verifiable traceability and context loss warnings | UC11 · UC25 |
| FR12 | Hybrid timeline and synchronized transcript editing | UC12 |
| FR15 | Explicit human approval and hallucination prevention | UC15 |
| FR18 | Audit logging and editorial decision tracking | UC22 |

# Non-functional requirements

### Performance & efficiency

**< 200 ms**

UI response while long videos are processed in the background

**≤ 15 min**

From a 1-hour recording to publishable clips

### Accuracy & reliability

**± 100 ms**

Word-level timestamps, linked to the original recording

**≥ 90%**

Speaker identification accuracy, up to 6 speakers

### Usability & accessibility

**< 300 ms**

Seek latency when editing from the transcript

**< 2 s**

Natural language search, ≥ 85% top-5 relevance

### Maintainability & modularity

**≥ 5**

Deployable modules, each replaceable on its own

**≥ 2**

LLM providers, swappable by configuration only

### Security, privacy & ethics

**0**

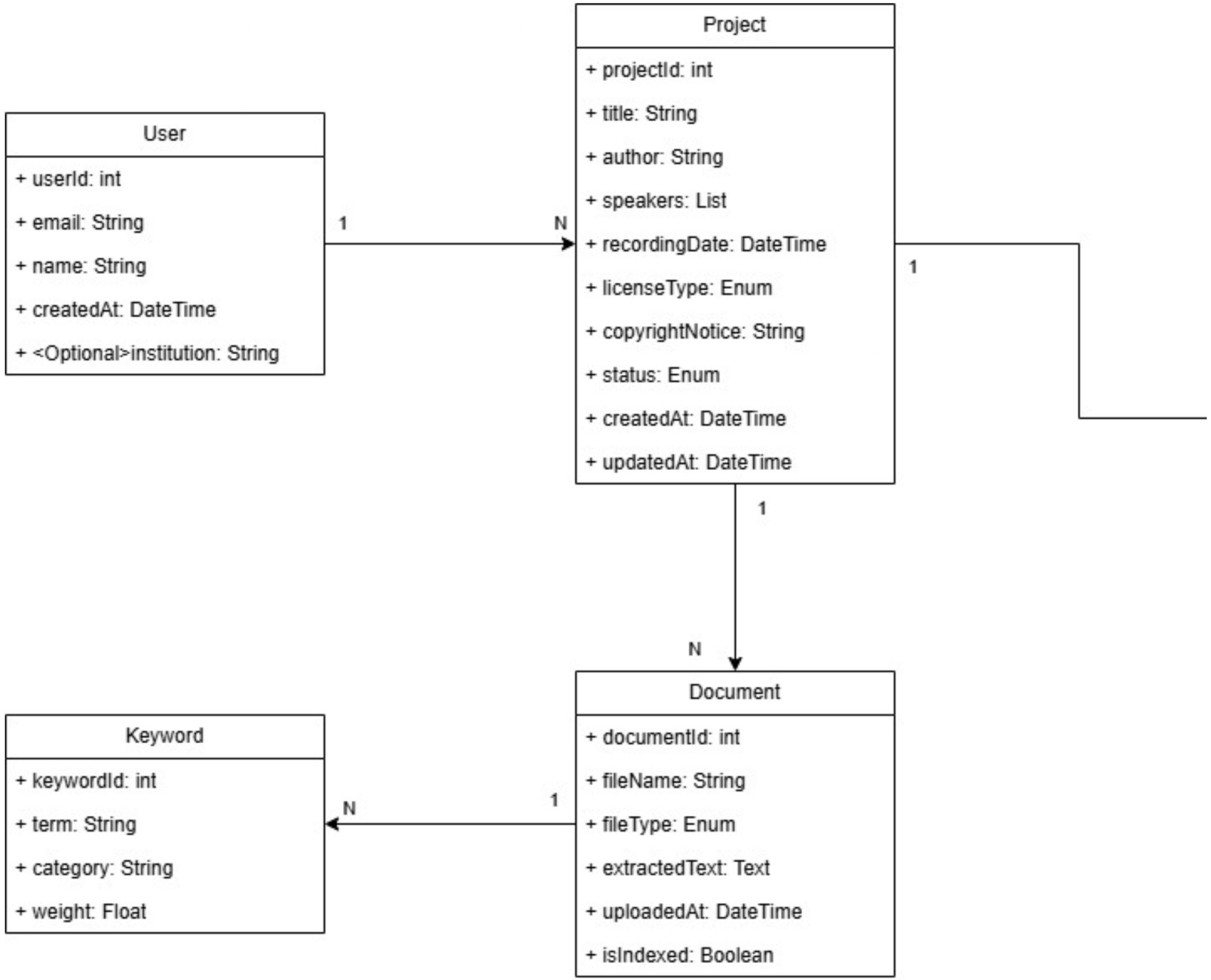
Altered quotes: every clip traces back to source timestamps

**AES-256**

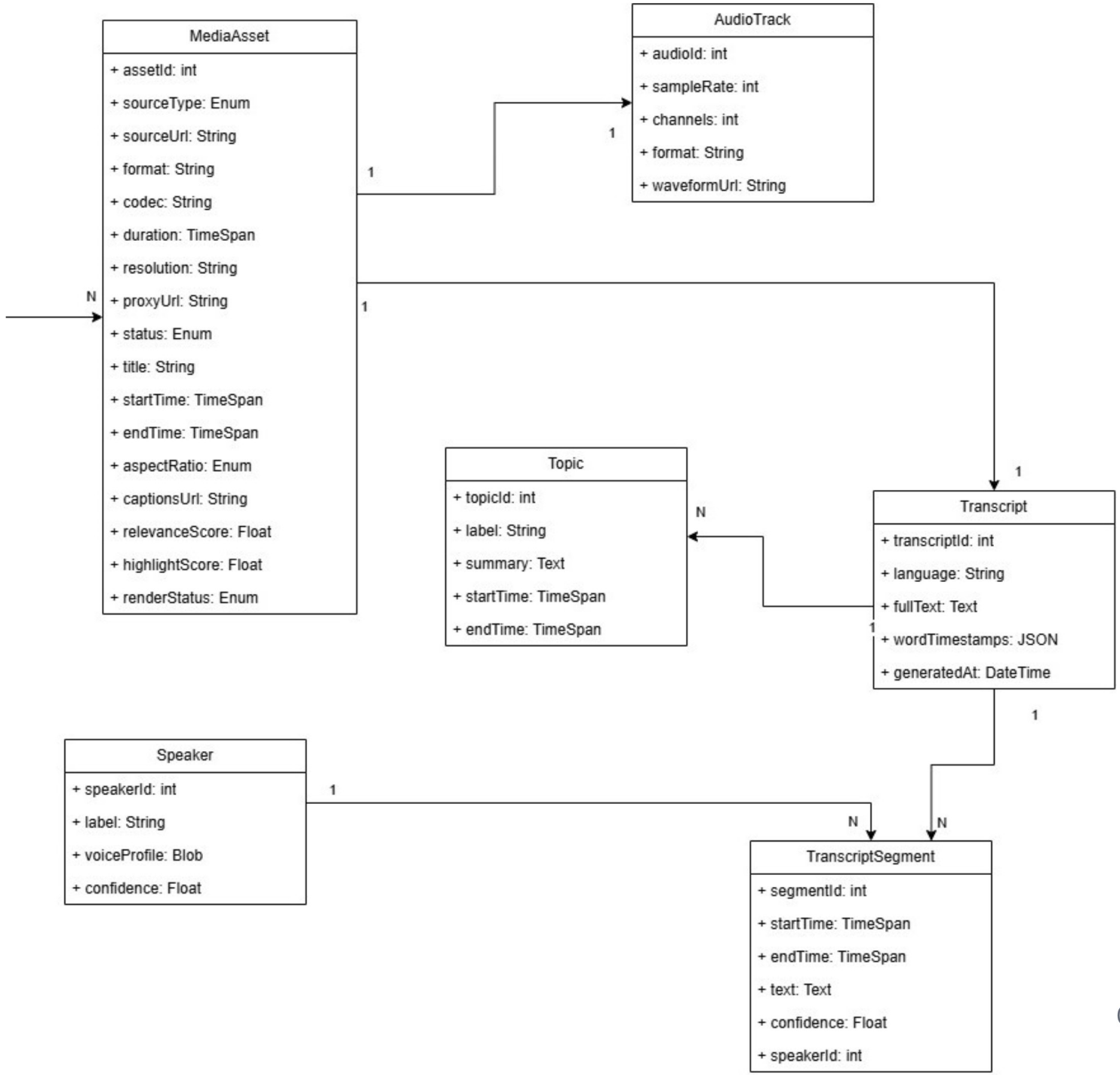
Data at rest, TLS 1.3 in transit, licence confirmed on every project

04 · DESIGN

# Domain model

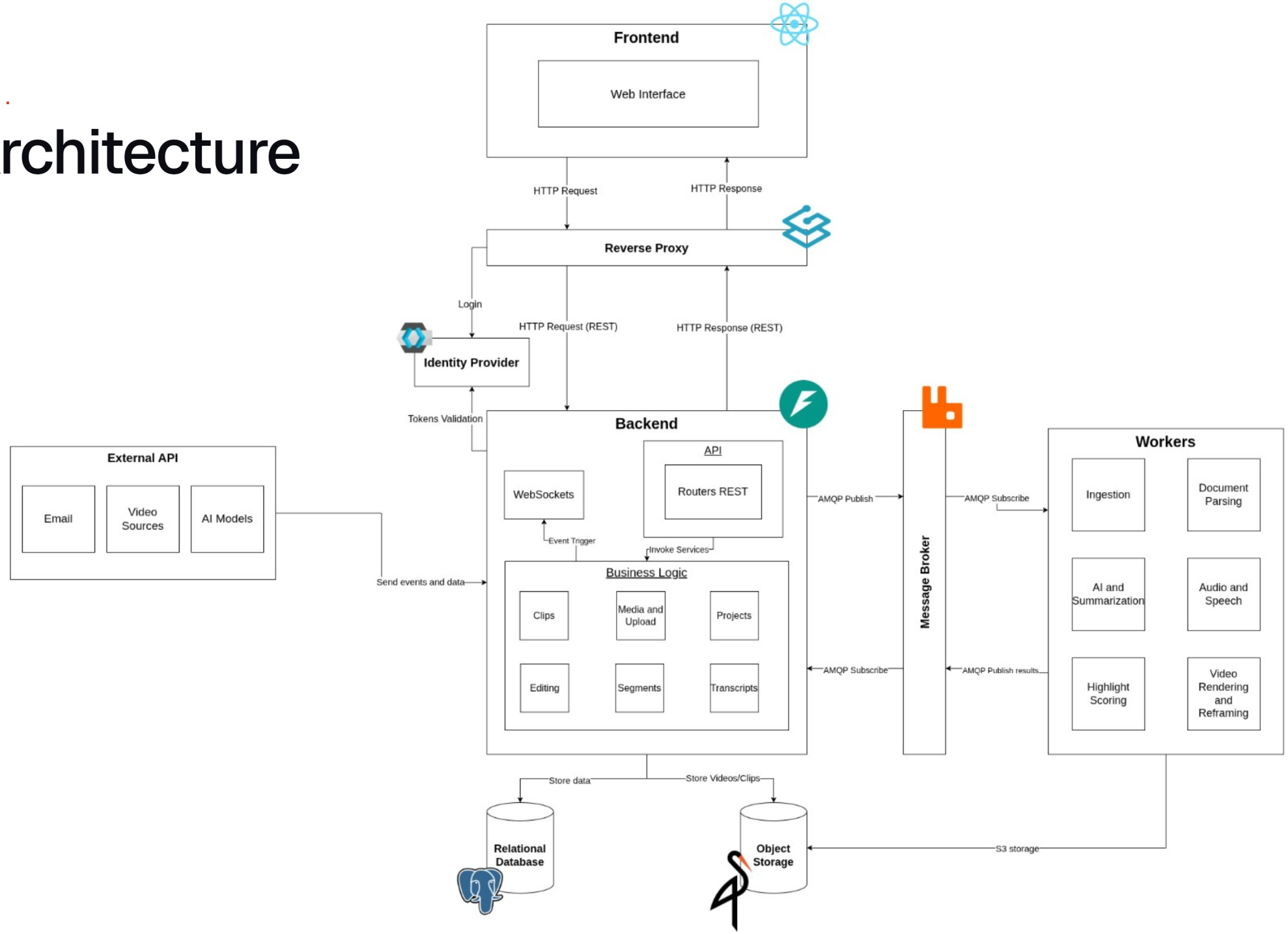


# Domain model



04 ·

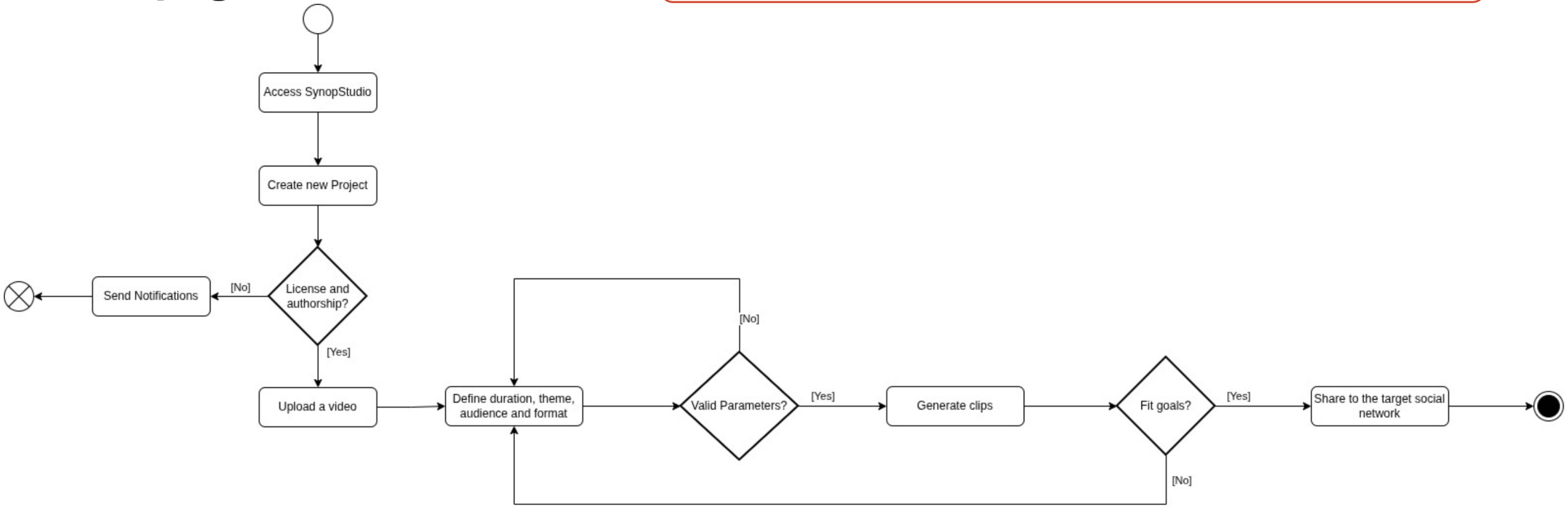
# Architecture



# Clip generation

UC09 · CONFIGURE CLIP GENERATION PARAMETERS

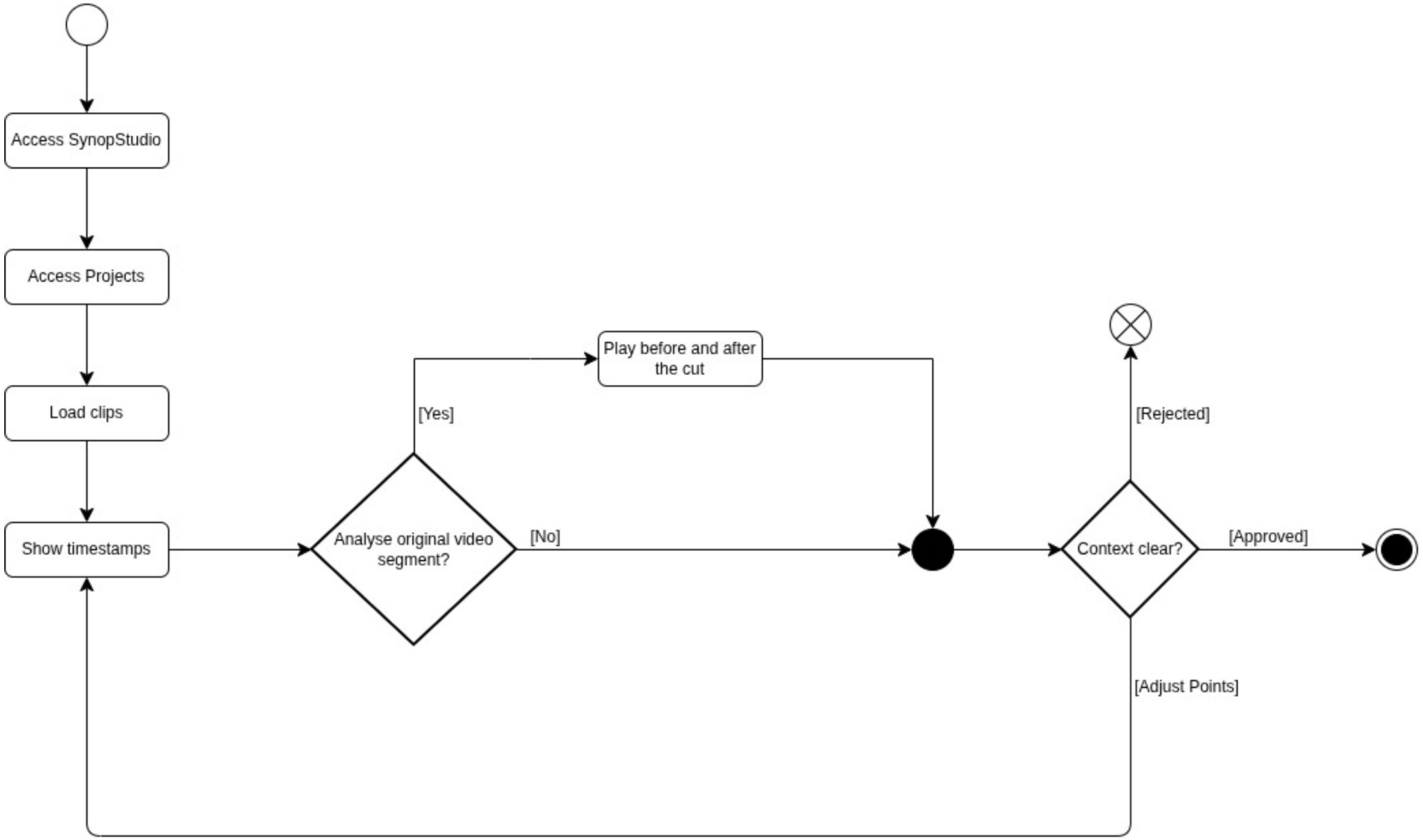
US15 · As a Content Creator, I want to set the theme, audience, duration and format before generating clips, so that the proposals fit my goal and the target social network.



# Clip review

UC11 · REVIEW CLIP PROPOSAL

US18 · As a Content Creator, I want each clip to show the intervals of the original video it was taken from, so that I can check the context before approving it.



# Thank you.

HTTPS://PEI-SYNOPSTUDIO.GITHUB.IO/DOCS/

